

CSE 604

Artificial Intelligence

Chapter 8: First Order Logic

Adapted from slides available in Russell & Norvig's textbook webpage

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Outline

- Why FOL?
- Syntax and semantics of FOL
- Using FOL
- Wumpus world in FOL
- Knowledge engineering in FOL

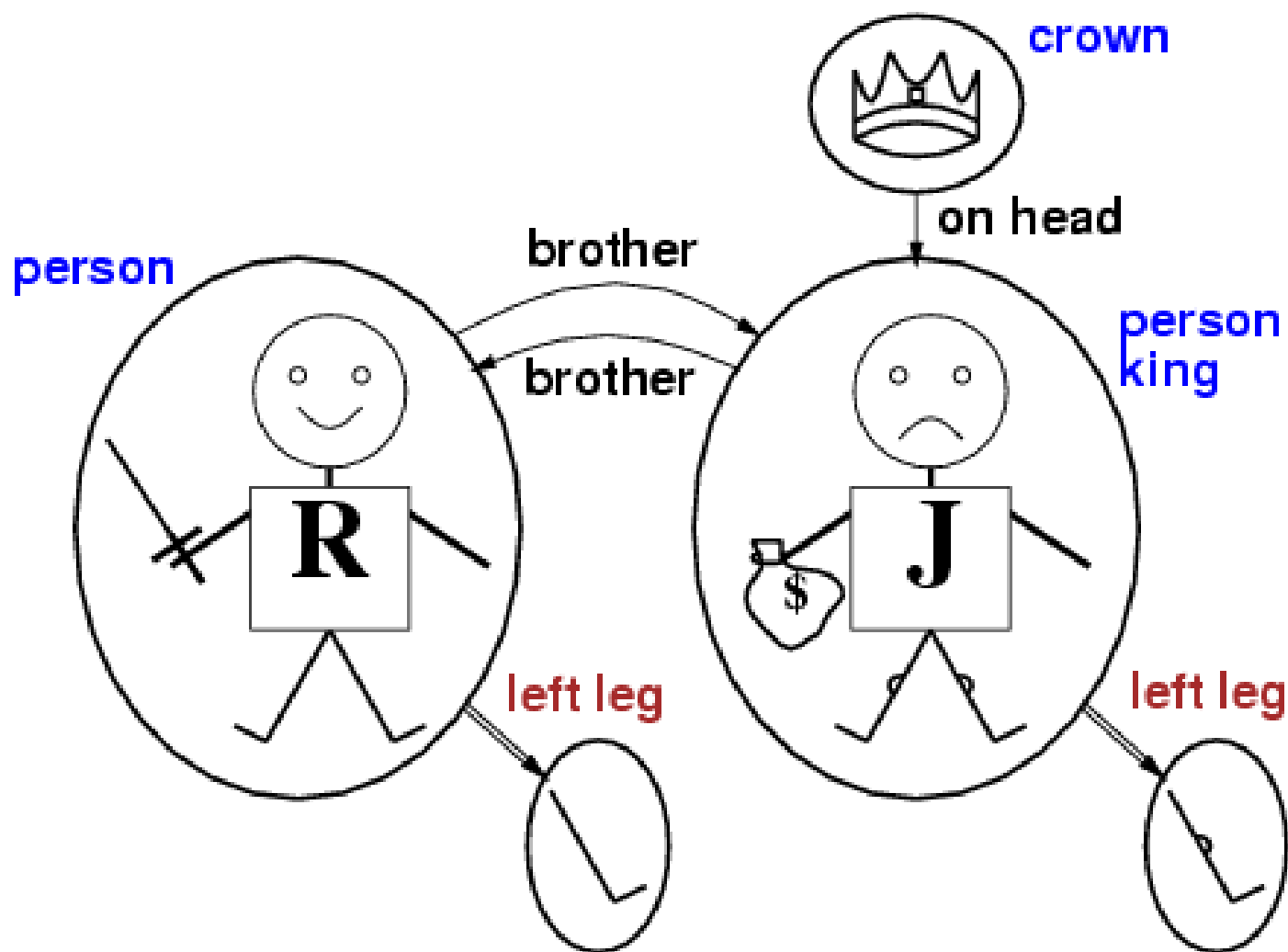
Pros and cons of propositional logic

- ☺ Propositional logic is **declarative**
 - Pieces of syntax correspond to fact
- ☺ Propositional logic **allows partial/disjunctive/negated information**
 - (unlike most data structures and databases)
- ☺ Propositional logic is **compositional**:
 - meaning of $B_{1,1} \wedge P_{1,2}$ is derived from meaning of $B_{1,1}$ and of $P_{1,2}$
- ☺ Meaning in propositional logic is **context-independent**
 - (unlike natural language, where meaning depends on context)
- ☹ Propositional logic has **very limited expressive power**
 - (unlike natural language)
 - E.g., cannot say "pits cause breezes in adjacent squares"
 - except by writing one sentence for each square

First-order logic

- Whereas propositional logic assumes the world contains **facts**,
- first-order logic (like natural language) assumes the world contains
 - **Objects**: people, houses, numbers, colors, baseball games, wars, ...
 - **Relations**: red, round, prime, brother of, bigger than, part of, comes between, ...
 - **Functions**: father of, best friend, one more than, plus, ...

Models for FOL: Example



Syntax of FOL: Basic elements

- Constants KingJohn, 2, IIT,...
- Predicates Brother, King, $>$,...
- Functions Sqrt, LeftLegOf,...
- Variables $x, y, a, b, \dots$
- Connectives $\neg, \Rightarrow, \wedge, \vee, \Leftrightarrow$
- Equality $=$
- Quantifiers $\forall, \exists$

Atomic sentences

Atomic sentence = *predicate* (*term*₁, ..., *term*_{*n*})
or *term*₁ = *term*₂

Term = *function* (*term*₁, ..., *term*_{*n*})
or *constant* or *variable*

- E.g., *Brother*(*KingJohn*, *RichardTheLionheart*)
- *>* (*Length*(*LeftLegOf*(*Richard*)), *Length*(*LeftLegOf*(*KingJohn*)))

Complex sentences

- Complex sentences are made from atomic sentences using connectives

$$\neg S, S_1 \wedge S_2, S_1 \vee S_2, S_1 \Rightarrow S_2, S_1 \Leftrightarrow S_2,$$

E.g. $Sibling(KingJohn, Richard) \Rightarrow Sibling(Richard, KingJohn)$

$$>(1,2) \vee \leq (1,2)$$

$$>(1,2) \wedge \neg >(1,2)$$

Truth in first-order logic

- Sentences are true with respect to a **model** and an **interpretation**
- Model contains objects (**domain elements**) and relations among them
- Interpretation specifies referents for
 - constant symbols** $\rightarrow$ **objects**
 - predicate symbols** $\rightarrow$ **relations**
 - function symbols** $\rightarrow$ **functional relations**
- An atomic sentence $predicate(term_1, \dots, term_n)$ is true iff the **objects** referred to by $term_1, \dots, term_n$ are in the **relation** referred to by $predicate$

Universal quantification

- $\forall \langle \text{variables} \rangle \langle \text{sentence} \rangle$

Everyone at IIT is smart:

$$\forall x \text{ At}(x, \text{IIT}) \Rightarrow \text{Smart}(x)$$

- $\forall x P$ is true in a model m iff P is true with x being each possible object in the model
- Roughly speaking, equivalent to the conjunction of instantiations of P

$$\begin{aligned} & \text{At}(\text{KingJohn}, \text{IIT}) \Rightarrow \text{Smart}(\text{KingJohn}) \\ \wedge & \text{At}(\text{Richard}, \text{IIT}) \Rightarrow \text{Smart}(\text{Richard}) \\ \wedge & \text{At}(\text{Pikachu}, \text{IIT}) \Rightarrow \text{Smart}(\text{Pikachu}) \\ \wedge & \dots \end{aligned}$$

A common mistake to avoid

- Typically, $\Rightarrow$ is the main connective with $\forall$
- **Common mistake:** using $\wedge$ as the main connective with $\forall$:

$$\forall x \text{ At}(x, \text{IIT}) \wedge \text{Smart}(x)$$

means “Everyone is at IIT and everyone is smart”!

Existential quantification

- $\exists \langle \textit{variables} \rangle \langle \textit{sentence} \rangle$
- Someone at CSE is smart:
 $\exists x \text{ At}(x, \text{CSE}) \wedge \text{Smart}(x)$
- $\exists x P$ is true in a model m iff P is true with x being some possible object in the model
- Roughly speaking, equivalent to the disjunction of instantiations of P
 - $\text{At}(\text{KingJohn}, \text{CSE}) \wedge \text{Smart}(\text{KingJohn})$
 - $\vee \text{ At}(\text{Richard}, \text{CSE}) \wedge \text{Smart}(\text{Richard})$
 - $\vee \text{ At}(\text{Pikachu}, \text{CSE}) \wedge \text{Smart}(\text{Pikachu})$
 - $\vee \dots$

Another common mistake to avoid

- Typically, $\wedge$ is the main connective with $\exists$
- **Common mistake:** using $\Rightarrow$ as the main connective with $\exists$:

$$\exists x \text{ At}(x, \text{IIT}) \Rightarrow \text{Smart}(x)$$

is true if there is anyone who is not at IIT!

Properties of quantifiers

- $\forall x \forall y$ is the same as $\forall y \forall x$
- $\exists x \exists y$ is the same as $\exists y \exists x$
- $\exists x \forall y$ is **not** the same as $\forall y \exists x$
- $\exists x \forall y \text{ Loves}(x,y)$
 - “There is a person who loves everyone in the world”
- $\forall y \exists x \text{ Loves}(x,y)$
 - “Everyone in the world is loved by at least one person”
- **Quantifier duality**: each can be expressed using the other
- $\forall x \text{ Likes}(x, \text{IceCream})$ $\neg \exists x \neg \text{Likes}(x, \text{IceCream})$
- $\exists x \text{ Likes}(x, \text{Broccoli})$ $\neg \forall x \neg \text{Likes}(x, \text{Broccoli})$

Fun with sentences

Brothers are siblings

$$\forall x,y \text{ Brother}(x,y) \Rightarrow \text{Sibling}(x,y)$$

“Sibling” is symmetric

$$\forall x,y \text{ Sibling}(x,y) \Leftrightarrow \text{Sibling}(y,x)$$

One's mother is one's female parent

$$\forall m,c \text{ Mother}(c) = m \Leftrightarrow (\text{Female}(m) \wedge \text{Parent}(m,c))$$

First cousin is a child of a parent's sibling

$$\forall x, y \text{ FirstCousin}(x, y) \Leftrightarrow \exists p, ps \text{ Parent}(p, x) \wedge \text{Sibling}(ps, p) \wedge \text{Parent}(ps, y)$$